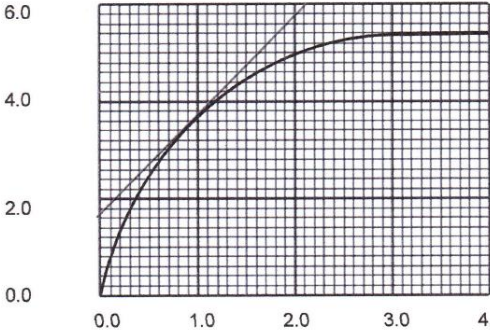


Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		Newton's 2 nd Law	1			1		
	(b)	(i)	Momentum /x10³ kgms⁻¹  0.0 2.0 4.0 6.0 0.0 1.0 2.0 3.0 4.0 time / s Suitable tangent at $t = 1.0 [\pm 0.1]$ s seen [$\Delta t \geq 1.0$ s] (1) Appropriate [with $\Delta t \geq 1.0$ s] values taken from tangent and manipulated correctly to show $F_{\text{resultant}} \approx 2\,000$ N (1) [ecf on tangent in range 1.7 – 2.3 kN]		2		2	2	
		(ii)	$m = 2\,000$ (or own value from (i)) $\div 0.4$ (= 5 000 kg)		1		1	1	
		(iii)	P labelled on line at $t \geq 3.0$ s	1			1		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	The vector sum of the momenta of bodies in a system stays constant (even if forces act between the bodies), (1)provided there is no external / resultant force / in an isolated system (1) Accept: The total momentum before a collision is equal to the total momentum after a collision..... (1)provided there is no external / resultant forces act / in an isolated system (1)	2			2		
		(ii)	Momentum before collision = 5.4×10^3 (Ns) (1) [from graph] Momentum after collision = (5 000 or ans to (b)(ii)+ 7 000) v (1) $v = 0.45$ (m s ⁻¹) (1) [ecf on value from graph, including slips in power of 10]	1	1 1		3	3	
			Question 5 total	5	5	0	10	6	0